Quiz 3 - Computational Physics I

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Date: Wednesday 30 April 2025 **Duration:** 45 minutes
Credits: 20 points (5 questions) **Type of evaluation:** LAB

Provide concise answers to the following items:

1. **(4 points) Numerical differentiation**
   (a) Indicate the sources of errors when computing the divergence of a 2D magnetic field: $\vec{\nabla} \cdot \vec{B}$
   (b) What defines the order of accuracy of a finite difference method?

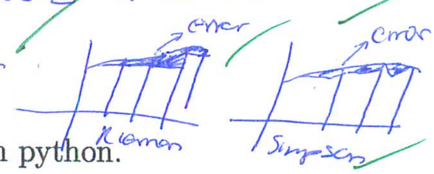
a) we could have a
machine epsilon error: commes from the limits of processing information in computer in terms of bits to allocate
method error: gradient function uses the finite difference method which can have errors of first $O(h)$ or second $O(h^2)$ grade

b) it is defined by the number of shifted points in the method which also gives us some error. Backward/Forward: First order
Central: Second order

2. **(4 points) Numerical integration**
   (a) Indicate 2 numerical methods used for calculating 1D integrals numerically.
   (b) Briefly explain how each method works.

Riemman Sum: This method calculates a rectangle bellow the curve per each dx and sums the area of the rectangles to approximate the total area or integral

Simpsons: It's an improvement of the Rieman sums, because a rectangle has some area left on the function edge, it uses a second order polinomial on the top insted to fit the function.



3. **(4 points) Root finding in python**
   (a) List 2 methods that we can use to find the roots of 1D functions in python.
   (b) Briefly explain how these python methods work to obtain the roots.

Newton-Raphson: This method takes 3 ~~things~~ steps/items/reference points, an x_0, $f(x_0)$, $f'(x_0)$, then it makes a guess on X_0 to create a tangent line to the point and compares with $n+1$ ~~and~~ the function to achive finding roots. Iterative until some criteria

$$n+1 = n - \frac{f(x)}{f'(x)}$$

Bisection: This method takes 2 points $f(x_0) > 0$, $f(x_1) < 0$ or viceversa to then calculate $f(\frac{x_0+x_1}{2})$ it checks whecher in which point was a sign change and does take that point and repeat iteratively until some tolerance

4. **(4 points) Fourier transforms**

(a) Provide a concise definition of the Fourier Transform and explain its significance for physics.

(b) Write down the mathematical formulae for the continuous and discrete Fourier Transforms.

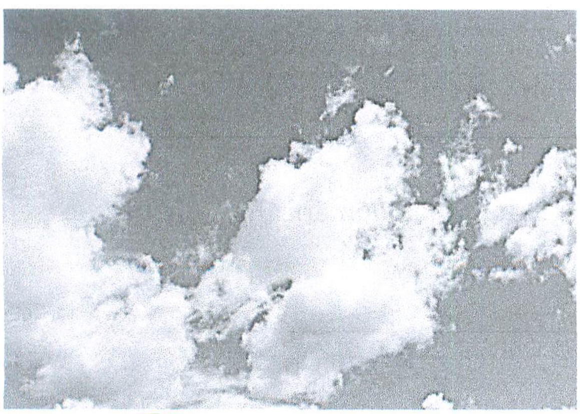
Fourier transform is a method to take some signal from Real Space (x) to the frequencies space (k). This is to be able to analize some characteristiques of the signal which can be done more easily on the FT space. Its composed of different sin or cos waves. Its important for physics because ALOT of things can be measure as a signal or spectra in physics, and normally comes with noise FT. helps us deal with this things easily

-0.25

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-i2\pi k x} f(x) dx \quad \Big/ \quad F(k) = \frac{1}{\sqrt{2\pi}} \sum_{n=0}^{\infty} A(x) \sin(x) + B(x) \cos(x)$$

X $\frac{1}{N}$ i

5. **(4 points) Image processing: edge detection**



Imagine you obtain the photograph (shown on the left-hand side) of atmospheric clouds, and you are asked to find and isolate the edges of the clouds. Design and sketch a suitable algorithm workflow to achieve this goal in python.

Re-Organise in functions and calls.

Start

I/O function?

-0.25

Start: Take the photo, open it and put the information into nd arrays

Inspect the data and plot the photo. using mat plot.lib

Using 2D FT, perform the FFT to K space and also shift the data to the center of the domain

Plot a histogram or spectra in the A x ~~?~~ plane to check the distribution and primarly characteristics of this signal

Define a High pass filter function

Define an step function to isolate the High frecuencys ~~de~~ the image

evaluate the signal on the High pass filter function.

calls

If the frecuency is inside the step function, zero it

if is out of the step function presere it

perform the inverse farier transform of the signal without ~~forgetting~~ to de-shift the signal

The edges should be detected since the edges on the images have a higher frecuency on R space

End.